MIL-E-1/1449A
28 March 1967
SUPERSEDING
MIL-E-1/1449
9 February 1962
(See note 2)

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8013A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Rectifier, high-vacuum, half-wave

See figure 1

Mounting position: Any, preferably vertical with base down

Weight: 2.9-ounce nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epx	Epp	ib	Io	tk	i(surge)
Unit:	Vac	kv	kVac	ma	mAdc	sec	ma
Maximum:	2.63	40	---	150	20	---	200
Minimum:	2.37	---	---	---	---	5.0	---
TEST CONDITIONS:	2.5	---	14	---	---	5.0	---

REQUIREMENT OR TEST:

GENERAL

3.2	Qualification - Required	E-50.3	Preheating
E-50.2	Holding period (t = 72 hours)	5.	Preparation for delivery

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 1							
Appendix D, 20(a), 30, 50	Visual and mechanical inspection criteria		---	---	---	---	---	---
1231	Peak emission	eb = 3,000 v (max) (see note 1)	0.65	II	is	450	---	ma
1256	Electrode current (anode)	Eb = 300 Vdc	0.65	II	Ib	33	59	mAdc
1301	Filament current		0.65	II	If	4.70	5.30	Aac
1353	Operation of rectifiers	ib = 100 ma (min) (see note 3)	0.65	II	Io	20	---	mAdc
	Quality conformance inspection, part 2							
Appendix D, 20(b)	Dimensions	See figure 1	6.5	L6	---	---	---	---
1036	Bump	Angle = 20°	6.5	L6	---	---	---	---
	Quality conformance inspection, part 3							
1501	Intermittent life-test operation	Group D; operation of rectifiers; ib = 150 ma (min); t = 500 hours	---	---	---	---	---	---
4.7.3	Life-test end point	Peak emission and failure to rectify	---	---	is	375	---	ma
	Periodic-check test (see 4.1.1.2)							
Appendix D, 20(b)	Dimensions	See figure 1	---	---	---	---	---	---

NOTES:

1. Voltage must be applied in pulses such that the tube will not be damaged.
2. Changes from previous issue. The extent of changes (deletions, additions, etc.) precludes the annotation of the individual changes from the previous issue of this document.
3. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - SH
Air Force - 11

Review activities:
Army - EL, MI
Navy - SH
Air Force - 11, 17, 85

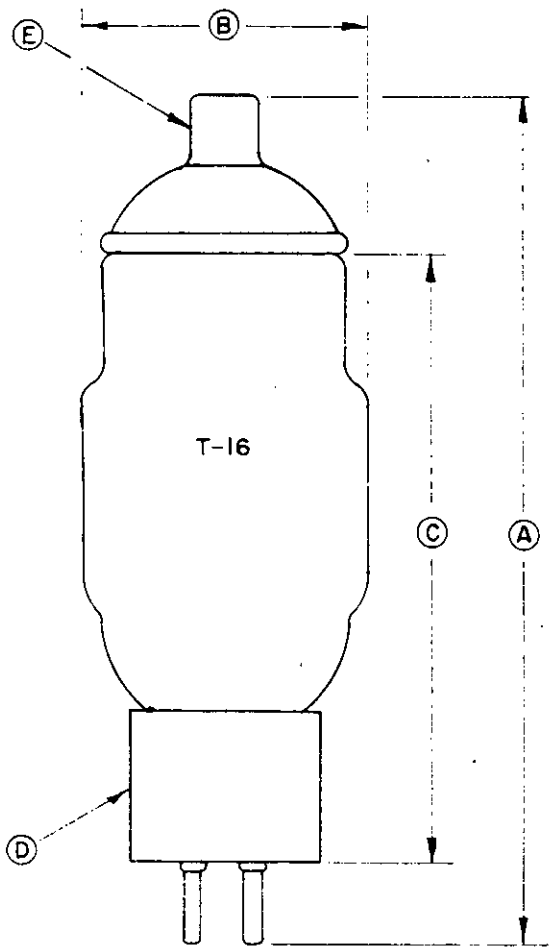
User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 14, 19

Preparing activity:
Navy - SH

(Project 5960-2247-89)

Code "C"

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PIN CONNECTIONS	
Pin No.	Element
1	f
2	nc
3	nc
4	f
cap	a

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note a)	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	5.75 (146.05)	6.07 (154.18)
B	---	2.07 (52.58)
C	---	4.507 (114.48)
Quality conformance inspection, part 3 (periodic check)		
D	Base: A4-9	
E	Cap: C1-19	

NOTE:
a. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Outline drawing of electron tube type 8013A.

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